ML Day 6

22 April 2026 06:12 PM

Hyperparameter tuning

→ Process of finding best parameter for ML model before training it. model gives best result

Cross validation → K=5 → 5 → K=1, K=2, K=3, K=4, K=5

K=5

Iteration 1

Train Data → (2, 20), (3, 30), (4, 40), (5, 50)

Test Data → (1, 10)

Iteration 2

Train Data → (1, 10), (3, 30), (4, 40), (5, 50)

5 Test Data → (2, 20)

Iteration 3

Train Data → (1, 10), (2, 20), (4, 40), (5, 50)

Test Data → (3, 30)

Iteration 4

Train Data → (1, 10), (2, 20), (3, 30), (5, 50)

Test Data → (4, 40)

Iteration 5

Train Data → (1, 10), (2, 20), (3, 30), (4, 40)

Test Data → (5, 50)

Cross val

Test Data → (5,50) — Cross val

k=5 → model will learn a dataset 5 times

k=1 → model will learn dataset for 1 time

train → Hyper tun → Cross val

test → act = pred ≠ ×

k=3 → three times → Best results

k=5 → five times → nearest values to Actual data

Data → k=1 → High variance (overfitting) → train → Good, test → Bad

k=5 → High Bias → train → Bad, test → Bad

k=3 → Good model [Balanced]

Split → RM, CM → Accurate results

→ try different k values on the dataset and selecting the one that gives best results

HP

Cross validation

Grid search → k=1, k=2, k=3, k=4, k=5 ---

Randomized search → k=3, k=6, k=2

k=5, k=7 ×

Data → DP → AFE → HPT → model Evaluation → New data → O/P

Data → DP → Alg → HPT → model Evaluation → New data → O/p

97% ✗ 80% 90% 100% 98% ✓ 92%

Linear relation

2 → 20
3 → 40
~~3.5 → 50~~
4 → 60
5 → 80
6 → 100

NLP ⇒ Natural Language Processing

↳ field of AI that helps Computers understand, read and process human language.

⇒ Chatbot
Google Translate

Hello → Machine

Hello → Telugu

Hello → ma → HL

Hy How are you.

text → Numbers

ML model understand only numbers

Clean, prepare the text

↳ Text preprocessing

↓

process of Cleaning and preparing text data before giving it to a ML model

"Hello!!! I am Learning Data SCIENCE!!!" ✓

Stop words

→ Capital letters

→ Punctuations

→ extra words (I, am)

→ punctuation
→ extra words (I, am)

→ Convert all text into small letters
↳ lowercasing

→ Remove Punctuation
"hello i am learning data science"

→ Remove stopwords → i, am, the, is, are etc
"hello learning datascience
↳ V3 ✗ → V1
→ learn

Convert words to base form → { Stemming / Lemmatization }

⇒ "hello learn data science" ✓ ✗ → machine
training words to machine
→ 1 num → n/u

i am ✗

Raw text → clean text
→ remove noise
→ reduce data size
→ makes data consistent